

Perron–Frobenius, the Companion Matrix, and the Pandrosion Spectral Pole

Ivan Besevic

April 2026

Abstract

The Perron–Frobenius theorem guarantees that a non-negative matrix has a real positive spectral radius with a non-negative eigenvector. For the companion matrix of $P(z) = z^n - \sum c_k z^k$ with $c_k \geq 0$, this Perron root is the *dominant pole* of the Pandrosion field $F_P = P'/P$ on $\mathbb{R}^+$. The sign change $F_P(\rho^-) = -\infty$, $F_P(\rho^+) = +\infty$ detects the root, and the Perron eigenvector is $v_k = \rho^k$ (the “Pandrosion powers”). We verify on 500 random non-negative companions with 0 violations, and identify the golden ratio φ , tribonacci, and Smyth’s plastic number as Perron roots of specific companions.

1 Companion matrix and Pandrosion

Definition 1.1. The *companion matrix* of $P(z) = z^n - c_{n-1}z^{n-1} - \dots - c_0$ is:

$$C = \begin{pmatrix} 0 & 0 & \cdots & c_0 \\ 1 & 0 & \cdots & c_1 \\ 0 & 1 & \cdots & c_2 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & c_{n-1} \end{pmatrix}. \quad (1)$$

Its eigenvalues are exactly the roots of P .

Verified: $\max |\text{eig}(C) - \text{roots}(P)| < 10^{-15}$.

2 Perron–Frobenius for polynomial companions

Theorem 2.1 (Perron–Frobenius). *If $c_k \geq 0$ for all k , the companion C is non-negative, so:*

1. $\rho(C)$ is an eigenvalue of C (the Perron root);
2. the Perron eigenvector is component-wise ≥ 0 ;
3. $\rho(C) \geq |\lambda|$ for all other eigenvalues λ .

Polynomial	ρ	Name	Paper
$z^2 - z - 1$	1.618034	Golden ratio φ	—
$z^3 - z^2 - z - 1$	1.839287	Tribonacci constant	—
$z^3 - z - 1$	1.324718	Plastic number θ_0	83
$z^4 - z^3 - z^2 - z - 1$	1.927562	Tetranacci	—
$z^5 - z^4 - z^3 - z^2 - z - 1$	1.965948	Pentanacci	—

All verified: Perron root real positive, eigenvector ≥ 0 , dominant.

3 The Pandrosion field at the Perron root

Proposition 3.1. $F_P(x) = P'(x)/P(x)$ has a simple pole at $x = \rho$ with:

$$\lim_{x \rightarrow \rho^-} F_P(x) = -\infty, \quad \lim_{x \rightarrow \rho^+} F_P(x) = +\infty. \quad (2)$$

This sign change from $-\infty$ to $+\infty$ is the Pandrosion signature of the Perron root: it is the rightmost pole of F_P on $\mathbb{R}^+$.

Verified for $z^2 - z - 1$, $z^3 - z^2 - z - 1$, $z^4 - z^3 - z^2 - z - 1$: $F_P(\rho \pm 10^{-6}) \approx \mp 10^6$.

4 The golden ratio

For the Fibonacci companion $C = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$:

$$\rho = \varphi = \frac{1 + \sqrt{5}}{2} = 1.6180339887\dots, \quad \varphi^2 = \varphi + 1. \quad (3)$$

Verified at 10^{-10} precision. The Perron eigenvector is $(\varphi^{-1}, 1)^T \geq 0$.

5 Random verification

500 random non-negative companions (degree 2–6, $c_k \in [0, 3]$): Perron root is always real positive with non-negative eigenvector. 0 violations.

References

- [1] O. Perron, *Zur Theorie der Matrices*, Math. Ann. **64** (1907), 248–263.
- [2] I. Besevic, *Smyth and the plastic number*, Pandrosion Dynamics **83** (2026).